

Whose Sentencing World?

Machine Learning, Integrated Gradients, and the Data Provenance Problem
in Japanese Criminal Judgments

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A large and largely unread body of text

- The reasoning section of a Japanese judgment — the *ryōkei no riyū* — is where judges set out the factors they weighed
- Japanese sentencing has been studied through legal doctrine and small case samples
- **To our knowledge, no published study had applied machine learning to Japanese criminal sentencing — until now**
- We read the corpus, and we read the corpus's own history

The computational sentencing landscape

- China: large-scale judgment prediction on public court corpora (Zhong et al. 2020)
- Europe: ECtHR outcome prediction and its critics (Medvedeva et al. 2020)
- New Zealand: interpretable sentencing models (Rodger, Lensen & Betkier 2023)
- Japanese legal NLP has grown in civil-law domains (Kondo et al. 2022; Yamada et al. 2024)
 - criminal sentencing remained untouched — partly a data-access story

Why theft is the right place to start

- The most common contested offence among elderly and socially marginal defendants
- At sentencing, the decisive question is often not what the person did but whether they have somewhere to return to
 - a guarantor, supervised housing, a welfare plan
- Japan has no public pre-sentence investigation system → welfare information reaches the court through the prose of the judgment itself

The corpus is not a neutral window

- Westlaw Japan over-represents contested and doctrinally noteworthy cases
 - the routine suspended sentence for a first-time elderly shoplifter — the statistical norm — is largely missing
- What is available for computation reflects an institutional Umwelt: written by judges for appellate readers, selected by editors for legal significance
- "Whose world, whose data?" is a measurable property of the material

Auditing what the database returned

- We set out to collect first-instance district-court decisions, 2010–2023
- The audit: judgments spanning 1958–2026 — only about three-fifths fall within the nominal window
 - plus a residue of appellate, summary- and family-court decisions
- How we audited: re-pulled the full bibliographic records, de-duplicated, and checked every date and court level against the metadata
- the slippage is itself part of the argument

Reading the audit as provenance, not as noise

- About a quarter of the judgments predate the intended window; roughly one in seven postdate it
- A small appellate residue — about three and a half percent — sits inside a nominally first-instance corpus
- Theft is the principal offence in about one case in six; about three in ten judgments carry no theft charge at all — the word appears only in the prose
- Our full-text query bounded the corpus by vocabulary — and by nothing else: not date, not instance, not legal category
- a query bounded by words met a collection bounded by editors

A deliberately lightweight, reproducible pipeline

- Reasoning sections extracted → tokenised with MeCab
- TF-IDF features + one structured field (prior-conviction count) → custodial vs. suspended sentence
 - Post-hoc audit (Sept 2026): an ‘age’ field used in earlier versions mostly records other people’s ages (victims, witnesses) in these anonymised judgments — withdrawn; linear results re-derived without it (text-only AUC 0.901; with prior convictions 0.925)
- L1-regularised logistic regression, ten-fold cross-validation → exact SHAP values
- No GPU needed · code, lexicon, CV splits and SHAP outputs packaged for release with the companion paper
 - raw text cannot be redistributed under Westlaw's licence — the retrieval procedure is documented for reconstruction

SHAP as distant reading

- Attribution answers a philologist's question: which words carry the decision?
- A concordance, weighted by consequence — every term ranked by its pull on the outcome
- The model is an instrument for discourse analysis, not a sentencing tool
 - nothing here is deployed on a person; we interrogate a vocabulary

The model recovers sentencing without being told any law

- Prior convictions push toward custody · settlement with the victim and the absence of a record push toward suspension
- Overall discrimination is high: AUC 0.90 from text alone, 0.93 with the prior-conviction count
- The prior-conviction count adds a little on top of the text (+0.02 AUC)
 - the prose largely, though not entirely, restates the case record

Welfare language sits on the decision boundary

- Fourteen of the fifty-nine welfare terms fall in the top tenth of the importance ranking — well above chance
- A complementary check — a fine-tuned Japanese BERT read with integrated gradients — surfaces the same welfare vocabulary at the aggregate level
- welfare talk is not decoration; it is part of what separates custody from suspension

...and it points both ways

- Toward leniency: "rehabilitation", "probation", "supervision" — 17 of the 26 lexicon terms the model selects
 - Toward incarceration: "social reintegration", "facility" — 9 terms, plausibly in statements of what is not available ("reintegration is difficult")
 - Judges reach for the language of welfare as a reason both to release and to confine
- direction = sign of the fitted weight; term-level readings and finer category profiles remain exploratory

When the two machines disagree

- The linear model and the neural model can disagree about an individual term
 - our first reading put "supervision" on the custodial side; the fitted weight and integrated gradients both put it on the suspension side — a reading error, caught by cross-checking
- A human word-sense check: only about six in ten lexicon hits are truly welfare-sense occurrences
- The main culprit is contextual polysemy — "society", "life", "care" inside idioms and compounds
- A separate trap — one reading, two words: kōsei is "rehabilitation" (更生) or the technical "constitute" (構成), one feature
- distant reading still needs philology

Robustness without heavy machinery

- Restrict to judgments where theft is the principal offence → BERT's discrimination essentially unchanged (the text-only linear model loses five points)
- Recluster the neural attribution at the judgment level → the signal holds in the audited folds
- Compare against frequency-matched non-welfare words → the enrichment survives
- cheap checks, chosen to attack our own reading

Reproducibility and licensing for legal DH

- The texts cannot travel: Westlaw's licence forbids redistribution
- What travels instead: retrieval documentation · code · lexicon · CV splits · SHAP outputs
- The disclosure package accompanies the companion paper — ask us for access
 - openness under licence constraints = publish the recipe, not the ingredients

What computation cannot see here

- No causal claims: we describe judicial language, not the effect of welfare on outcomes
- The corpus is the published, contested world — not routine practice
- Word-sense noise caps how precisely the lexicon can be read
- Finer-grained patterns are exploratory until they replicate
 - the audit names the limits of the claims; it does not launder them

Three points for digital humanities

- 1. Japanese criminal judgments can be opened as a computational corpus — carrying legal DH beyond its civil-law, anglophone centre of gravity
 - 2. Explainable AI works as an instrument of discourse analysis: not whether welfare language appears, but how it behaves
 - 3. Data provenance becomes a working variable, not a closing disclaimer
- a stance that carries over to any project built on institutionally curated collections — our own included

Coda: whose sentencing world?

- The world of published, contested cases — assembled by editors, written for appellate readers
- We measured that gap and made it part of the argument
- within that world, welfare discourse is structurally embedded in how judges write about punishment

Thank you

- Whose sentencing world? — the world of published, contested cases; we measured the gap and made it part of the argument
- Openness: code · lexicon · CV splits · SHAP outputs — with the companion paper (retrieval documented)
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